CLAUDE CODE HACKATHON · GATES FOUNDATION × ANTHROPIC

Reclass Intake

A reclass request today is a free-text email that takes a month to fail. This is that email as a form that knows the rules.

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Team reclass-intake · Gates Foundation × Anthropic · 14 August 2026

16 business rules · 389 tests · 4 agents in parallel · 0 cloud resources · runs on localhost

THE PROBLEM

An error costs a round trip, not a keystroke.

Every error is a round trip

The request arrives as prose. Someone interprets it against a rulebook that disagrees with itself and writes back. The requester re-reads it, re-pulls the transaction and resubmits. Every step there is a person.

The rules live in one head

Every request funnelled to one senior manager — “the alias, like the Wizard of Oz.” Four written sources that don't agree: even whether the threshold is “greater than” or “greater than or equal” is contradictory.

Requesters can't self-serve

The rules aren't discoverable at the moment of submission, so people learn them by being rejected. The foundation's own improvement document asked for hover tips. Nobody built it.

THE PROCESSING CYCLE — FROM THE GUIDANCE, NOT AN ESTIMATE

You submit
Any weekday. Nothing validates it.

Processed
Last business day of the month.

Visible downstream
10th business day of the next month.

A malformed request isn't caught until the batch — and the correction goes into the next cycle.

WHAT CHANGES

The email becomes a structured request.

BEFORE	AFTER
Free-text email	**Structured diff: this transaction, these fields, these values**
Requester types a cost center that can't receive charges	**It isn't selectable**
Error caught at month-end batch	**Error caught before submit**
Rules in one head + four disagreeing documents	**Rules in version-controlled code, with citations and tests**
Requester learns by rejection	**Requester learns by tooltip**
Submissions reassembled by hand for handoff	**One consolidated file, generated**
“When will this land?”	**“Processes 31 Aug · posts by 14 Sep”**

Same requester, same inbox, same downstream file. The difference is that the request is now checkable.

WHAT IT DOES

Paste a key. Pick the changes. Get the file.

1 Paste a GL Business Key

One field, unique. No compound-key ambiguity.

2 The transaction arrives live

23 fields from the Budget-to-Actuals model. No export, no copy-paste.

3 Tick what changes

Cost center, account, project/event, contract ID — any combination.

4 Pick new values

Dropdowns from live dimension data, your division first. Invalid targets visible but unselectable.

5 16 rules run as you type

Each citing the document it came from. Where sources disagree, it says so.

6 Export the Working List

One workbook for accounting, in the format the process already consumes.

Nothing is sent automatically. The tool produces a file and an audit record — a human still hands it off.

THE ONE THING TO REMEMBER

You can't pick an invalid cost center.

Because it isn't selectable — and it's still shown, so you learn why.

NEW COST CENTER

10420	Enterprise Business Intelligence
10530	Central Finance
12300	Global Health Program Services
~~40005~~	~~(aggregation-only rollup node)~~

40005 looks real.

It's an aggregation-only rollup node. No dollars are ever coded to it, so a reclass targeting it can never post.

Shown, not hidden.

A requester who expected to find it learns why it isn't available. Hiding it would teach nothing.

Live data proved this out: the dimension query only returns cost centers that carry dollars, so 40005 never came back at all. Marking only what returned would have silently dropped it.

Two things the process asked for.

When will this land?

The form answers it at submission.

Processes Mon 31 Aug · visible by Mon 14 Sep

Reclasses process on the last business day of the month and are visible downstream by the 10th business day of the next. Inside two business days of the cutoff, the form says so.

Holidays and the December–January year-end cycles are flagged, not guessed at.

I know. Submitting anyway.

Judgment stays with the human — on the record.

- [ ] I know this violates the reclass submission guidelines and am submitting as an exception

Four rules can be acknowledged: threshold, prior year, already-reclassed, budget-only cost center. The finding drops to INFO with the requester's name on it.

Eight never can, including payroll. No tick box makes a missing transaction exist, or moves more money than the transaction holds.

Four agents in parallel, against a frozen contract.

CONTRACT LOCK · FIRST 15 MINUTES
Shared types, fixtures and a stub API surface — committed before any track started.

A Power BI

Live lookup by GL Business Key and dimension dropdowns. Owns the network, so it owns the offline path.

B Rules

16 rules as pure functions over frozen fixtures, with the tests that prove them.

C Frontend

The entire UI, built against stub endpoints that existed from minute zero.

D Export

Working List XLSX, batch consolidation, and the audit log.

A hook, not a convention.

.claude/hooks/contract_lock.py is a PreToolUse hook that blocks edits to the frozen contracts. Tested against 32 cases. Mid-sprint it stopped an agent that had legitimate authorisation — and that agent escalated to a human instead of routing around it.

STAYING POWER

The repo ships the machinery that built it.

`.claude/agents/`	The build crew, committed. Four subagent definitions declaring exclusive files, forbidden paths and the command that proves them done. Sprint 1 was four calls in one message.
`.claude/hooks/contract_lock.py`	Enforcement, not convention. Blocks edits to the frozen contracts and the vendored Power BI connector.
`.claude/skills/add-rule/`	/add-rule "<plain english rule>" scaffolds the function, a test, a fixture and the citation. Rule 17 gets added by someone who isn't the author.
`scripts/pre_commit_check.py`	Blocks secrets and live financial data from entering the repo — and blocks --no-verify.

The gate grew teeth.

It caught a real incident: three genuine vendor rows with real contract IDs and amounts reached a demo fixture. A later review found the scanner itself read nine files as empty on Windows and reported success. Fixed — then extended with a second checker that asserts data invariants, with tests that break each one to prove the guard still bites.

We were wrong three times, and wrote it down.

We invented a story about the business.

We designed a multi-field reclass as one atomic row. It demoed well. The process owner corrected it on sight: accounting wants one Working List row per change, each carrying its instruction in column X — the field they actually read, which we had left blank.

Export, rule and README rewritten; the citation records who corrected it.

A fresh clone was unusable.

Our documentation told people to set a variable in .env to get offline mode. Nothing in the codebase loaded .env. The file was inert, the default attempted an interactive Power BI sign-in, and the dropdowns hung.

The default is now offline. A fresh clone works with no configuration at all.

The export stated the wrong amount.

Column X carried no figure, so a partial reclass told accounting to move the entire transaction. Six of the eight rows in our own demo file were partials — most of the artifact, not an edge case.

Found by an independent review, not by us.

Each was found by someone reading the work critically. None was quietly patched.

Pilot it without changing anything downstream.

Same file, same process

The output is the Working List format accounting already consumes. No migration, no new system of record. It slots in front of the existing process rather than replacing it.

Offline from a fresh clone

Two commands, no credentials, no Power BI access, no cloud resource provisioned at any point. A colleague can try it this afternoon.

It feeds the automation behind it

Every submission writes a structured, labelled audit record. The four weakest categories in the existing email-triage automation are exactly what this form validates.

NOT BUILT — AND SAID SO

Redistribution is designed, not built. · “Budget-only cost center” has no agreed definition — we tested the obvious heuristic against the ledger, it was wrong, and the rule ships dormant rather than guessing. · Delegation-of-authority validation needs a data source we don't have. · ~2,800 lines outside the rule engine are still untested.

BY THE NUMBERS

One day, two sprints, zero cloud resources.

Build time	**One day, two sprints**
Business rules	**16, each citing its source**
Tests	**389**
Agents run in parallel	**4 in Sprint 1, 9 across the day**
Cloud resources provisioned	**0**
Real production data committed	**0 — hook-enforced, test-verified**
Setup for a new user	**2 commands, no credentials**

START HERE

`backend/rules.py`
16 rules, each citing its source

`.claude/`
the hook, the agents, the skill

`README.md`
two commands, no credentials

`docs/EXECUTIVE-SUMMARY.md`
the written submission

The zeroes are the load-bearing ones: nothing was provisioned, and no real financial data can enter the repository.